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Problem

FROM TASK TO WORK

- Current agents can search or copy files but struggle with presentations, reports and cross-application workflows.
- Two bottlenecks dominate: precise GUI visual grounding and long-horizon cognitive understanding.
- Behavior-only data omits why humans choose an action, recover from errors or revise a plan.

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PC Tracker

DATA COLLECTION

- A lightweight background tool records mouse/keyboard actions, screenshots and accessibility information.
- Task-oriented and natural collection modes capture exploration, pauses, corrections and cross-app behavior.
- Privacy controls, transparent data storage and refinement remove tracker actions and redundant events.

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Cognition completion

TURN CLICKS INTO REASONING

- Stage 1 converts click coordinates into concise semantic descriptions using marked screenshots and element metadata.
- Stage 2 reconstructs the thought behind each action from task, past steps, future actions and visual state.
- The result is a cognitive trajectory containing progress awareness, exploration and error recovery.

PC AGENT

PC Agent: While You Sleep, AI Works

A Cognitive Journey into the Digital World | 2024

Transfer human cognition - not just clicks - by collecting GUI trajectories, reconstructing action meaning and thought, then training planning and grounding agents.

GUI agent

cognition transfer

multi-agent

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PC Agent

PLANNING + GROUNDING

- A Qwen2-VL planning agent generates interleaved thoughts and actions from cognitive trajectories.
- A Molmo grounding agent converts click descriptions to coordinates and self-validates with GUI metadata.
- Invalid targets trigger feedback and re-planning; PyAutoGUI executes accepted actions.

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Evidence

SMALL DATA, LONG WORK

- Only 133 trajectories fine-tune Qwen2-VL-72B; guided presentation traces average about 50 events.
- The agent completes practical multi-page presentations with sequences up to 50 steps across Chrome and PowerPoint.
- Batch poster creation: 11 of 20 outputs meet requirements (55% success).

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Meaning & limits

TAKEAWAY

- High-quality cognitive data appears far more data-efficient than raw behavior cloning for complex digital work.
- Failures are hard to recover from; performance depends strongly on detailed task descriptions.
- Generalization, drag/scroll actions, long-term planning, privacy and evaluation of subjective deliverables remain open.